

JOHNSON MUGARRA

Data Scientist | AI/ML Specialist | Research & Data

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PROFESSIONAL SUMMARY

Data scientist and research consultant with 6+ years building machine-learning solutions, predictive-analytics systems, and automated data pipelines across health, education, gender-based violence and SRHR, agriculture, climate, and household livelihoods. Work spans the full ML lifecycle alongside monitoring and evaluation, interactive BI dashboards, and rigorous data-quality frameworks. Co-author on peer-reviewed agronomy research, and comfortable translating complex analysis for technical teams and executive audiences alike.

CORE TECHNICAL SKILLS

Programming & ML Python, R, SQL, Java · TensorFlow, PyTorch, Scikit-Learn, Statsmodels, Prophet, XGBoost, Keras

Methods Time-series forecasting, classification & regression, clustering & segmentation, causal inference, NLP (sentiment & network analysis), computer vision

Visualisation & BI Power BI (Microsoft Certified), Tableau (Certified Specialist), Looker Studio, Plotly, D3.js

Data & MLOps ETL pipelines, Airflow, Spark, Hadoop, Docker, Git, PostgreSQL, MongoDB, SurveyCTO, API integration, model monitoring & retraining

Cloud AWS, Azure, Google Cloud

PROFESSIONAL EXPERIENCE

Research & Data Consultant

Nov 2025 - Present

Enimiro (vanilla & coffee agribusiness) · Remote

- Consult on and put in place the farmer data system: seven automated pipelines covering SurveyCTO-to-PostgreSQL ETL, NASA POWER climate data, Prophet harvest forecasts, purchase-loyalty and pollination analysis, and inspection follow-ups.
- Mentor and guide the management team to run the system themselves, so the capability stays in-house.
- Developed a random-forest vanilla yield-estimation model over 275 surveyed farmers (14 farm features), cutting mean absolute error from a 158 kg median baseline to 101 kg on out-of-fold validation.
- Led standalone studies: a vanilla-farmer Living Income study (212 farmers, Anker/Fairtrade benchmarks) and the Kayunga agrohub needs assessment; produced interactive and static district plot maps (GeoPandas, Folium).
- Authored concept notes on data quality, field accountability, and zonal vanilla production for management.

Research & MEAL Specialist

Jan 2025 - Present

Center for Research & Communication (CRC) · Kampala, Uganda

- Lead the data and analytics behind research and evaluation consultancies for UN agencies and international NGOs, including IRC, UNICEF, World Vision, Save the Children, Danish Refugee Council, Oxfam, GIZ, and ADRA, across Uganda, Somalia, and the Horn of Africa. The portfolio spans health, education and teaching practice, child protection, gender-based violence and SRHR, food security, and livelihoods.
- Run impact and endline evaluations, KAP surveys, and needs assessments end to end: tool design, SurveyCTO data collection, data-quality assurance, analysis, and the final reports clients deliver to donors.
- Write the technical proposals for new evaluation bids, and run workshops on ML, statistical modelling, Python, R, and Power BI.
- Recent delivery includes the IRC MERP endline survey funded by the German Federal Foreign Office.

Data Scientist

Jan 2024 - May 2025

MAAIF · VANEX · Makerere University · Catholic Relief Services

- Designed and deployed an end-to-end forecasting system integrating API, satellite (NASA POWER), and ground data, cutting mean absolute error by 18% against the previous model.
- Led the full ML lifecycle - data collection, model development, validation, deployment, and monitoring - with automated retraining protocols.

- Built and automated scalable ETL pipelines with daily API synchronisation using Python and Airflow, consolidating data into a central warehouse.
- Developed interactive Tableau/Power BI dashboards giving non-technical stakeholders real-time access to insights.
- Presented findings and policy recommendations at a national forum attended by H.E. President Yoweri Kaguta Museveni (April 2025).

Research Associate

Jun 2023 - May 2025

USAID-SITES · SSS Inc. · DLH Corporation

- Ran programme evaluation and auditing across USAID-funded health projects in Uganda, including President's Malaria Initiative (PMI) activities and the DHIS2 and electronic medical records (EMR) transition for HIV care.
- Conducted facility readiness and data-quality assessments across hospitals in the Eastern, Central, and Northern regions of Uganda, working alongside Makerere University and the Ministry of Health.
- Supervised digital data collection and ran SQL/Python data-quality audits, achieving 95%+ data reliability, with stakeholder dashboards and automated reporting for programme teams.
- Engineered ETL pipelines across SurveyCTO, Power BI, and PostgreSQL with daily API automation.

Data Analyst / Research Associate

Jul 2021 - Jun 2023

Bronkar (U) Limited (research & M&E consultancy) · Kampala, Uganda

- Designed M&E frameworks and integrated digital data-collection tools that reduced field-operations time by 25%.
- Ran mixed-methods statistical analysis and produced evidence-based recommendations for clients across East Africa.
- Led capacity building and staff training on analytics tools, building durable data skills in client teams.

SELECTED PROJECTS

Vanilla Yield Estimation Model (Enimiro)

2026

- Random-forest model predicting recorded harvest for 275 farmers; beat the median baseline (MAE 101 kg vs 158 kg) with certification status, land size, and plant count as the strongest predictors.

Vanilla Farmer Living Income Study (Enimiro)

2026

- Quantified the gap between farmer earnings and a living-income benchmark using Anker and Fairtrade reference values; a modular Python/R pipeline cleaned 485 survey columns from 212 farmers into an analysis-ready dataset and written report.

Automated Impact-Evaluation Pipeline

2023 - 2026

- Python + Airflow pipeline with automated ingestion, causal-inference modelling, and auto-generated reports, cutting evaluation cycles from six weeks to under ten days.

Social-Media Sentiment & Network Analysis (MSc capstone, MAHE)

2024

- Analysed social-media data for sentiment, and modelled the network of users and interactions to study how information flows.

PUBLICATIONS

- Obua, T., Sserumaga, J.P., Tukamuhabwa, P., Namara, M., Awio, B., Mugarra, J., Tusiime, G., & Chigeza, G. (2024). Unravelling Yield and Yield-Related Traits in Soybean Using GGE Biplot and Path Analysis. *Agronomy*, 14(12), 2826. <https://doi.org/10.3390/agronomy14122826>
- Co-author (2025). Identifying Optimal Lines for Enhanced Symbiotic Performance in a Mini-Core Collection of Cowpea [*Vigna unguiculata* (L.) Walp]. *Advances in Agriculture*. <https://doi.org/10.1155/aia/2202496>

EDUCATION

M.Sc. Data Science

Sep 2022 - Oct 2024

Manipal Academy of Higher Education (MAHE) · Karnataka, India

- Coursework: Machine Learning, Predictive Analytics, Time-Series Forecasting, NLP, Big Data Analytics, Advanced Regression, Statistical Modelling, Data Visualisation.

B.Sc. Agricultural Science

Sep 2016 - May 2021

Makerere University · Kampala, Uganda

- Coursework: Biometrics & Statistics, Research Methods, Mathematics.

CERTIFICATIONS

- Google Project Management Certificate
- Microsoft Certified: Power BI Data Analyst Associate
- SAS Certified Statistical Business Analyst
- Tableau Desktop Specialist

RECOGNITION

- Selected to present analytics research findings and policy recommendations at a national forum attended by H.E. President Yoweri Kaguta Museveni, at Makerere University (April 2025).